APPENDIX

A. The detailed decomposition process

From the am-gm inequality, it is easy to know that (8.5) has a solution when C in (9.1) is fixed. And the larger C is, the better the solution of (8.5). Since the negotiation breakdown solution is of constant value, C can take the maximum value after the optimal solution of the social resilience enhancement subproblem (SP1) is obtained.

$$C = \underbrace{(C_{\mathrm{PDS}}^{\mathrm{op,NO}} + C_{\mathrm{GDS}}^{\mathrm{op,NO}})}_{\text{The negotiation breakdown solution}} - \underbrace{(C_{\mathrm{PDS}}^{\mathrm{op}} + C_{\mathrm{GDS}}^{\mathrm{op}})}_{\text{The solution in SP1}} \tag{9.1}$$

Since the multi-energy support cost is offset in SP1. The Nash bargaining method is introduced to determine t equilibrium transaction settlement rule to calculate and allocate the multi-energy support cost. The natural logarithm function is a strictly monotonically increasing convex function. Taking the logarithm of (8.5) and converting the primal problem into a minimum problem. So, (8.5) is equivalent to SP2 when the solution of SP1 is determined.

B. The case data

TABLE IV
THE COST COEFFICIENTS

Cost coefficients	Value (10^4\$)	Cost coefficient	Value (10^4\$)
$\ell_{eb}^{\mathrm{shedding}}$	$\begin{cases} 1.943 & t\in[1,7] \\ 3.997 & t\in[8,13]\cup[17,18]\cup[22,24] \\ 6.493 & t\in[14,16]\cup[19,21] \end{cases}$	$\ell_{gb}^{\mathrm{shedding}}$	0.021
$\ell_{el}^{\mathrm{switch}}$	0.1	$\ell_{gl}^{\mathrm{switch}}$	0.1
$\ell_{eb}^{\mathrm{inter}}$	$\begin{cases} 0.117 & t\in[1,7] \\ 0.24 & t\in[8,13]\cup[17,18]\cup[22,24] \\ 0.39 & t\in[14,16]\cup[19,21] \end{cases}$	$\ell_{gb}^{\mathrm{inter}}$	0.001
$\ell_{em}^{\mathrm{crew}}$	0.05	$\ell_{gm}^{\mathrm{crew}}$	0.05

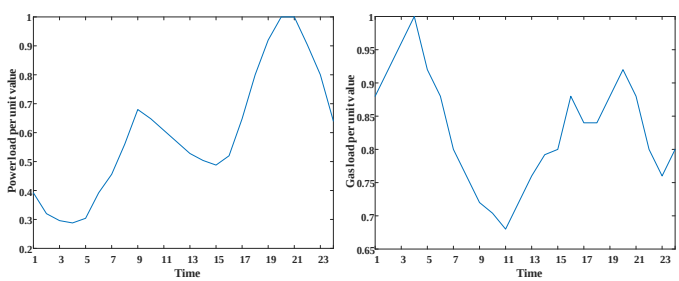

Fig. 10 The load curve

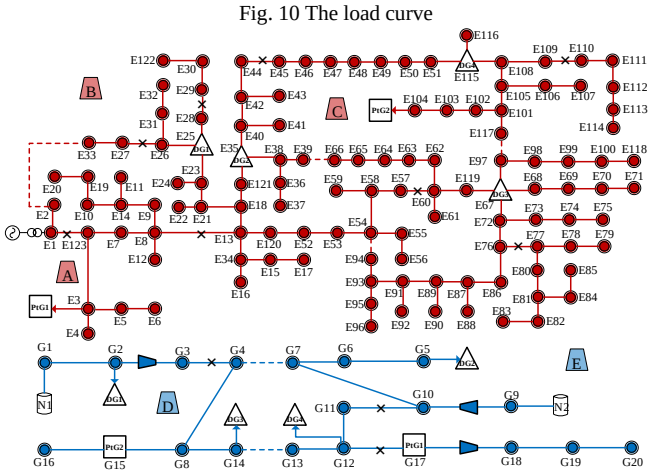

Fig. 11 Diagram of modified 123-20-buses EGDS